FORMAL EFFECTIVENESS (APPROXIMATION EQUIVALENCE) FOR THE STACK OF COHERENT SHEAVES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1506. Standing situation; S locally Noetherian, f separated.
- Equation artin-equation-approximation (canonical tag 07X6). cited at statement line 1509. The functor being studied.
- Lemma artin-lemma-effective (canonical tag 07X8). cited at proof line 1521. Reduction to fixed Spec(A) -> B.
- Theorem spaces-more-morphisms-theorem-grothendieck-existence (canonical tag 08BE). cited at proof line 1528. Grothendieck existence.
- Remark spaces-more-morphisms-remark-reformulate-existence-theorem (canonical tag 08BF). cited at proof line 1536. Reformulation remark.
- Lemma spaces-more-morphisms-lemma-flatness-over-Noetherian-ring (canonical tag 08VP). cited at proof line 1542. Flatness criterion.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line: the lemma is phrased in this situation; load-bearing.

Source reference. Original equation 07X6: sources/stacks-chow/tags.txt, source lines 10066–10066. Cited at statement line 1509: the functor shown to be an equivalence.

Source reference. Original lemma 07X8: sources/stacks-chow/tags.txt, source lines 10068–10068. Cited at proof line 1521: reduce to objects over a fixed map to B.

Source reference. Original theorem 08BE: sources/stacks-chow/tags.txt, source lines 10564–10564. Cited at proof line 1528: coherent modules = compatible systems.

Source reference. Original remark 08BF: sources/stacks-chow/tags.txt, source lines 10565–10565. Cited at proof line 1536: the explicit equivalence.

Source reference. Original lemma 08VP: sources/stacks-chow/tags.txt, source lines 11242–11242. Cited at proof line 1542: flatness criterion.

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Lemma 0.2. *In Situation 08KB assume that S is a locally Noetherian scheme and that $f : X \to B$ is separated. Let $\mathcal{X} = \mathcal{C}oh_{X/B}$. Then the functor Artin's Axioms, Equation (07X6) is an equivalence.*

Proof. Let A be an S-algebra which is a complete local Noetherian ring with maximal ideal $\mathfrak{m}$ whose residue field k is of finite type over S. We have to show that the category of objects over A is equivalent to the category of formal objects over A. Since we know this holds for the category $\mathcal{S}_B$ fibred in sets associated to B by Artin's Axioms, Lemma 07X8, it suffices to prove this for those objects lying over a given morphism $\mathrm{Spec}(A) \to B$.

Set $X_A = \mathrm{Spec}(A) \times_B X$ and $X_n = \mathrm{Spec}(A/\mathfrak{m}^n) \times_B X$. By Grothendieck's existence theorem (More on Morphisms of Spaces, Theorem 08BE) we see that the category of coherent modules $\mathcal{F}$ on X_A with support proper over $\mathrm{Spec}(A)$ is equivalent to the category of systems $(\mathcal{F}_n)$ of coherent modules $\mathcal{F}_n$ on X_n with support proper over $\mathrm{Spec}(A/\mathfrak{m}^n)$. The equivalence sends $\mathcal{F}$ to the system $(\mathcal{F} \otimes_A A/\mathfrak{m}^n)$. See discussion in More on Morphisms of Spaces, Remark 08BF. To finish the proof of the lemma, it suffices to show that $\mathcal{F}$ is flat over A if and only if all $\mathcal{F} \otimes_A A/\mathfrak{m}^n$ are flat over $A/\mathfrak{m}^n$. This follows from More on Morphisms of Spaces, Lemma 08VP. □